Martine Lokken

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EMPLOYMENT

Postdoctoral Researcher 02/11/2023–
Incl. Juan de la Cierva Postdoctoral Fellowship, 01/01/2026-
Institut de Física d'Altes Energies, Bellaterra (Barcelona), Spain

Graduate research and teaching assistant 01/09/2018–31/08/2023
David A. Dunlap Department of Astronomy & Astrophysics, University of Toronto, Canada

EDUCATION

PhD, University of Toronto, Canada 01/09/2018–07/11/2023
Thesis: *The Interconnected Cosmos: A Study of Multi-Scale Correlations in the Late Universe*
David A. Dunlap Department of Astronomy & Astrophysics
Advisors: Prof. J. Richard Bond and Prof. Renée Hložek

BS, University of Virginia, USA 25/08/2014–30/05/2018
Incl. semester abroad, University of Edinburgh, Fall 2016
Astronomy-Physics, Highest Distinction

RESEARCH INTERESTS

Beyond- ΛCDM cosmology, alternative statistics for large-scale structure, cosmic web evolution via CMB × LSS cross-correlations, impact of feedback on the intergalactic medium, Lyman-α forest cosmology

PUBLICATIONS WITH KEY ROLE

Total published papers: 10 in lead author list, of which 6 first-author; 30 other
Citations: As first author: 62, as contributing author: 6,241 (Source: NASA ADS)

In lead author list *(Contributions described in italics when not first-author)*

1. Rosenberg**,** I., **ML**, Hložek, R., et al. (2026). *Here, There and Everywhere: How AGN jets affect galaxy cluster environments*. Submitted to A&A. arXiv:2607.13132 *Developed core science idea and plan, co-supervisor throughout project, writing and analysis for Sec 4.3, 4.5*
2. **ML,** Bond, J. R., Hložek, R., et al. (2026). *The superclustering of hot gas: cosmological sensitivity in the Websky simulations*. JCAP. arXiv:2510.08331
3. Casas, L., Herrera-Alcantar, H. K., Chaves-Montero, J., et al. incl. **ML** (2026). *Validation of the DESI DR2 Lyα BAO analysis using synthetic datasets*. PRD. *Tested the impact of damped Lyman-α systems on the BAO measurement to inform baseline analysis configuration.* arXiv: 2503.14741
4. **ML**, Font-Ribera, A., & McDonald, P. (2025). *Weighted FFT estimators for 1D and 3D correlations of the Lyman-α forest*. JCAP. arXiv:2505.02904
5. **ML**, van Engelen, A., Aguena, M., et al. (2025). *Superclustering with the Atacama Cosmology Telescope and Dark Energy Survey. II. Anisotropic Large-scale Coherence in Hot Gas, Galaxies, and Dark Matter*. ApJ. arXiv:2409.04535
6. Chaves-Montero, J., Cabayol-Garcia, L., **ML**, et al. (2025). *ForestFlow: predicting the Lyman-α forest clustering from linear to nonlinear scales*. A&A. *Modeling of the P-cross statistic with emulator and validation of predictions with measurements on hydro simulations, Sec 6.3* arXiv:2409.05682v2
7. Anbajagane, D., Chang, C., Baxter, E. J., et al. incl. **ML** (2024). *Cosmological shocks around galaxy clusters: a coherent investigation with DES, SPT, and ACT*. MNRAS. *Computed large-scale orientations and contributed to their interpretation for Sec. 5.* arXiv:2310.00059
8. **ML**, Cui, W., Bond, J. R., et al. (2023). *Boundless baryons: how diffuse gas contributes to anisotropic tSZ signal around simulated Three Hundred clusters*. MNRAS. arXiv:2211.00242

9. **ML**, Gagliano, A., Narayan, G., et al. (2023). *The simulated catalogue of optical transients and correlated hosts (SCOTCH)*. MNRAS. arXiv: 2206.02815
10. **ML**, Hložek, R., van Engelen, A., et al. (2022). *Superclustering with the Atacama Cosmology Telescope and Dark Energy Survey. I. Evidence for Thermal Energy Anisotropy Using Oriented Stacking*. ApJ. arXiv:2107.05523
11. Fritz, T. K., **ML**, Kallivayalil, N., et al. (2018). *The Orbit and Origin of the Ultra-faint Dwarf Galaxy Segue 1*. ApJ. *Corrected data for distortions, wrote Sec 3.5, 3.6.* arXiv: 1711.09097

SELECTED PRESENTATIONS

Durham University, UK. DESI Summer Collaboration meeting: plenary — 15/07/2026
"3D small-scale correlations of the Lyα forest with DESI" (with S. Satyavolu)

IFCA, Santander, Spain. Cosmology seminar series: **invited seminar** — 06/05/2026
"Shaping the cosmic web: how cosmology and AGN feedback affect the late-time distribution of hot gas"

ASU School of Earth and Space Exploration, USA: seminar — 18/12/2025
"How cosmology and AGN feedback affect the late-time distribution of hot gas"

IFCA, Santander, Spain. XI Meeting for Fundamental Cosmology: **invited plenary** — 19/11/2025
"Cosmology with the DESI Lyman-α forest"

Sesto, Italy. Cosmology on the Steep Rise: conference talk — 6/02/2025
"New tools for extracting cosmology from the small-scale Lyman-α forest"

S'Agaro, Spain. DES Collaboration Meeting: **invited plenary** — 29/05/2024
"Anisotropic large scale coherence in hot gas, galaxies, and dark matter from ACTxDES"

KICP, University of Chicago, USA. Cosmo seminar series: **invited seminar** — 23/02/2023
"Boundless Baryons: the anisotropic superclustering of cosmic gas"

KICC, Cambridge, UK. New Physics from Old Light: workshop talk — 16/09/2024
"*Anisotropic relationships in the cosmic web: from ACTxDES to ACTxDESI"*

Rio de Janeiro, Brazil. Cosmo '22: conference talk — 2022
"Aligning the Cosmic Web: Superclustering at the intersection of ACT+DES data and simulations."

FELLOWSHIPS

Juan de la Cierva postdoc. fellowship (*National, Spain,* **35,700 EUR/yr**) — 01/01/2026-01/01/2028
NSERC PGS-Doctoral (*National, Canada,* **21,000 CAD/yr**) — 01/09/2021-12/09/2023
QEII Grad. Schol. in Science and Tech. (*Provincial*, **10,000 CAD/yr**) — 13/07/2020-01/09/2021
CITA Entrance Scholarship (*Departmental graduate award*, **5,000 CAD**) — 01/09/2018-30/08/2019
NSF Grad. Research Fellowship Program — Offered 01/04/2018
National, USA, **34,000 USD/yr**, *declined to attend program in Canada*

SUPERVISION

PhD: Lucía Flores Lopez, *Maximizing the small-scale information from the Lyman-α forest* — 2025-
Universitat Autonoma de Barcelona, co-supervised w. Andreu Font Ribera

Undergraduate thesis: Marta Doz Cárceles, *Probing AGN jet feedback through radio galaxy stack* — 2026
Universitat Autonoma de Barcelona, co-supervised w. Tony Mroczowski

Undergraduate thesis: Isaac Rosenberg *Feedback variations in The300 Cluster simulations* — 2024-2025
University of Toronto, co-supervised w. Renée Hložek

TEACHING

Teaching Fundamentals Certificate 2023
Received certification through course from Teaching Assistants' Training Program, U of T

Directed Reading Program Lead 2022-2023
Planned and supervised cosmology-focused reading + discussion group with 3 undergraduates

Teaching Assistant U of Toronto Dept. of Astronomy & Astrophysics undergraduate courses
Prepared and led tutorials, managed email and discussion boards, marked and invigilated exams

Astronomy at the Frontier (121)	Spring 2023
Stars and Galaxies (201)	Spring 2019, Summer 2020, Spring 2022
Galaxies and Cosmology (222)	Spring 2021
Origin and Evolution of the Universe (121)	Spring 2020
Programming Support TA, serving various U of T astronomy courses	Fall 2020
The Sun and its Neighbors (101)	Fall 2018, Fall 2019

LEADERSHIP AND SERVICE

Topical group co-lead, DESI Lyα small-scale clustering	2025-
Hub meeting lead organizer, DESI Meetings Committee	2025-
Founder and co-organizer, IFAE women's lunches	2024-
PhD student mentor, DESI and IFAE mentoring programs	2023-
Paper referee, MNRAS, ApJ, Open Research Europe; 7 internal reviews for DESI, DES, LSST	2023-
Seminar series co-organizer, IFAE, CITA	2022-2026
Workshop co-organizer, DESI Lyman-α Barcelona hub	2025
Sustainability Report co-author, DESI: collected collab-wide input + delivered plenary	2024-2025
Peer mentorship coordinator, U of T Astro department	2019-2020, 2021-2022
Sustainability white paper co-author, Can. Astr. Soc. sustainability committee	2020-2022
Conference co-organizer, CASCA Annual General Meeting	2022

COMPUTATIONAL SKILLS

Languages: Python (*advanced*); C++ (*proficient)*; Fortran, HTML, Julia (*basic)*
Skills: Parallelization with MPI and OpenMP, sci. computing on HPC clusters (SLURM, HTCondor, PBS, TORQUE systems), N-body and hydro simulation post-processing, Bayesian parameter inference with minimization and MCMC sampling (emcee, iminuit), collaborative code development
Research software contributions (GitHub repos): marlena6/oriented_superclustering; igmhub/cupix; igmhub/ForestFlow; LSSTDESC/transient-host-sims; ACTCollaboration/DR6_notebooks

SELECTED OUTREACH

Platform Dalí Barcelona, Spain *Science + art public interviews and discussion program*	2026
Cientifiques Lleida, Spain *Presentation to elementary school students*	2025
Bojos per la Física High school coding/physics workshop	2024, 2025
Visions of Science, Pursue STEM Toronto, CA *High school astro workshops*	2021, 2023
U of T AstroTours Co-director, Planetarium director, Public speaker	2018-2022

COLLABORATIONS

Simons Observatory (2026-) cluster working group, **Dark Energy Spectroscopic Instrument** (2023-): DR2 Lya-Px analysis lead, ACTxDESI filaments analysis lead, **Atacama Cosmology Telescope** (2019-): Led several tSZ x LSS projects, DR6 web content and tutorial notebook contributor, **LSST Dark Energy Science Collaboration** (2019-2023): Co-led transient-host simulations, core ELAsTiCC Challenge member, **The Three Hundred Project** (2022-): Led study of beyond-halo gas, oversaw new feedback-variation cluster simulations